

Forecasting Violent Crime in Chicago Community Areas with Machine Learning

Nuo Chen

Lian Duan

Zhe Gao

Yunxuan Liu

Wenlan Xie

Department of Economics
University of Chicago

May 29, 2026



- ① Introduction
- ② Data
- ③ Methods
- ④ Results
- ⑤ Selected References

- 1 Introduction
- 2 Data
- 3 Methods
- 4 Results
- 5 Selected References

The Question

By how much do regularised linear and tree-based machine-learning methods improve out-of-sample forecasts of monthly violent crime rates across Chicago community areas relative to autoregressive panel OLS benchmarks, and which feature groups account for the predictive gain?

- Predictive, *not* causal: lower out-of-sample MSE is the only objective.
- Panel of **77 community areas** $\times$ **168 months** (2010–2023); **408,040** violent incidents; strict no-leakage timing rule.
- Four families: **panel OLS, Random Forest, XGBoost, LSTM**.
- **Headline:** XGBoost on the OLS feature set wins (RMSE 3.786); *more features do not help*, and the OLS benchmark is hard to beat.

- 1 Introduction
- 2 Data
Sources and Descriptive Statistics
- 3 Methods
- 4 Results
- 5 Selected References

Outcome: violent crime rate per 10,000 residents

$$Y_{i,t} = \frac{C_{i,t}}{\text{Pop}_i} \times 10,000.$$

Seven public sources:

Dataset	Source
Violent crime incidents	Chicago Data Portal / CPD
311 vacant building	Chicago Data Portal, two endpoints
311 graffiti	Chicago Data Portal
Weather	NOAA GHCN-D, O'Hare USW00094846
ACS 5-year demographics	U.S. Census Bureau / CMAP
Community-area boundaries	Chicago Data Portal shapefile
Transit stops	CTA GTFS feed

Training and Holdout Samples

Variable	Train 2010–2019		Holdout 2021–2023	
	Mean	SD	Mean	SD
Crime rate $Y_{i,t}$	8.93	8.53	9.95	9.19
Vacant 311 rate $V_{i,t}^{311}$	2.68	4.59	2.63	4.24
Spatial crime lag $(WY_t)_i$	9.12	5.81	10.00	6.11
Avg. temperature (°F)	51.73	18.49	55.92	16.83
Precipitation (inches)	7.22	4.31	5.99	3.90

- Training: $N = 9,009$ community-area months after dropping initial lags.
- Holdout: $N = 2,541$ community-area months, evaluated once.
- Mean violent crime rate rises by about 11% after the pandemic break.

1 Introduction

2 Data

3 Methods

- Empirical Framework

- Panel OLS Benchmarks

- Tree-Based Machine Learning

- Recurrent Neural Network

4 Results

5 Selected References

Purpose	Period	Use
Training and tuning	2010–2019	Expanding-window CV
Excluded break period	2020	COVID structural break
Final holdout test	2021–2023	One-time evaluation

Expanding-window CV: train $\leq 2015 \rightarrow$ validation 2016; ...; train $\leq 2018 \rightarrow$ validation 2019.

Holdout metrics, $|\mathcal{T}| = 2,541$

$$\text{RMSE} = \sqrt{\frac{1}{|\mathcal{T}|} \sum_{(i,t) \in \mathcal{T}} (Y_{i,t+1} - \hat{Y}_{i,t+1})^2}.$$
$$R_{\text{OOS}}^2 = 1 - \frac{\sum_{(i,t) \in \mathcal{T}} (Y_{i,t+1} - \hat{Y}_{i,t+1})^2}{\sum_{(i,t) \in \mathcal{T}} (Y_{i,t+1} - \bar{Y}_{\text{train}})^2}.$$

No-leakage rule: all predictors for $Y_{i,t+1}$ are dated at or before month t .

Benchmark 1: autoregressive panel with fixed effects

$$Y_{i,t+1} = \alpha_i + \gamma_{m(t+1)} + \rho_0 Y_{i,t} + \rho_1 Y_{i,t-1} + \rho_2 Y_{i,t-2} + u_{i,t+1}.$$

Benchmark 2: + lagged weather and disorder

$$Y_{i,t+1} = \alpha_i + \gamma_{m(t+1)} + \boldsymbol{\rho}^\top L_{i,t} + \beta_1 T_{t-1} + \beta_2 P_{t-1} + \beta_3 S_{t-1} + \theta V_{i,t-1}^{311} + u_{i,t+1}.$$

- α_i absorbs persistent community-area differences.
- $\gamma_{m(t+1)}$ absorbs month-of-year seasonality.
- Lagged crime rates capture short-run persistence.

Spatial Durbin-style OLS: + queen-contiguity neighbour lag

$$Y_{i,t+1} = \alpha_i + \gamma_{m(t+1)} + \boldsymbol{\rho}^\top L_{i,t} + \lambda_0(WY_t)_i + \lambda_1(WY_{t-1})_i \\ + \beta_1 T_{t-1} + \beta_2 P_{t-1} + \beta_3 S_{t-1} + \theta_0 V_{i,t-1}^{311} + \theta_1(WV_{t-1}^{311})_i + u_{i,t+1}.$$

- W is the row-normalised queen-contiguity matrix.
- $(WY_t)_i$ captures neighbouring-area crime exposure.
- $(WV_{t-1}^{311})_i$ captures neighbouring disorder exposure.

Purpose

Tree-based models test whether nonlinear interactions improve forecasts beyond the linear AR + FE benchmark.

Random Forest

$$\hat{f}_{\text{RF}}(z) = \frac{1}{B} \sum_{b=1}^B T_b(z; \Theta_b, \mathcal{D}_b).$$

- Averages many decorrelated trees.
- Reduces variance relative to one deep tree.
- Uses six nested feature sets: RF1 $\rightarrow$ RF6.

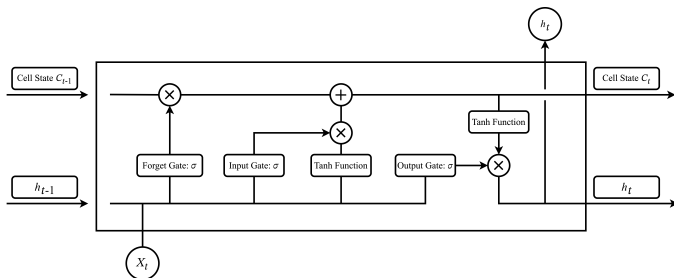
XGBoost

$$\hat{f}_{\text{XGB}}(z) = \sum_{m=1}^M \eta h_m(z).$$

- Adds shallow trees sequentially.
- Uses shrinkage and regularisation.
- Compares OLS features with an expanded feature pool.

Key test: if XGBoost beats spatial OLS using the same variables, the gain comes from nonlinear functional form.

LSTM with 3-Month Lookback



Architecture

- Two stacked LSTM layers.
- Hidden size: 256.
- Dropout: 0.20.
- AdamW, $\eta = 10^{-3}$.

Input sequence

- 3-month lookahead.
- Crime lags and rolling means.
- 311 vacancy measures.
- Month and area indicators.

Interpretation

- Forget gate keeps useful memory.
- Input gate adds new signals.
- Output gate forms the prediction signal.

Purpose: test whether recurrent sequence pooling adds value beyond hand-built lags and fixed effects.

- 1 Introduction
- 2 Data
- 3 Methods
- 4 Results**
- 5 Selected References

Unified Holdout Comparison, 2021–2023

Model	Features	RMSE	MAE	OOS R^2
<i>Panel OLS</i>				
B1: AR + community FE + month FE	91	3.853	2.298	0.836
B2: + weather + 311	95	3.837	2.293	0.837
Spatial Durbin-style	98	3.822	2.295	0.838
<i>Random Forest</i>				
RF1 crime history	3	4.149	2.434	0.796
RF5 + community dummies	95	3.821	2.217	0.827
RF6 + weather	100	3.828	2.234	0.826
<i>XGBoost</i>				
XGB on OLS-equivalent features	99	3.786	2.248	0.830
XGB expanded 269-feature pool	269	3.803	2.249	0.829
<i>LSTM</i>	134	3.966	2.404	0.810

XGBoost on the same feature set as the spatial OLS gives the lowest RMSE.

	B1 (AR+FE)	B2 (+W+311)	Spatial
$Y_{i,t}$	0.277***	0.266***	0.212***
$Y_{i,t-1}$	0.165***	0.149***	0.115***
$Y_{i,t-2}$	0.019	0.009	-0.013
T_{t-1}		0.064***	0.042***
P_{t-1}		-0.027**	-0.015
$V_{i,t-1}^{311}$		0.081**	0.055*
$(WY_t)_i$			0.313***
$(WY_{t-1})_i$			0.031
Community FE / month FE	Yes / Yes	Yes / Yes	Yes / Yes
Observations	9,009	9,009	9,009

- Persistence is concentrated in the most recent one to two months.
- The contemporaneous spatial lag is highly significant.
- Adding $(WY_t)_i$ lowers the own-area lag coefficients.

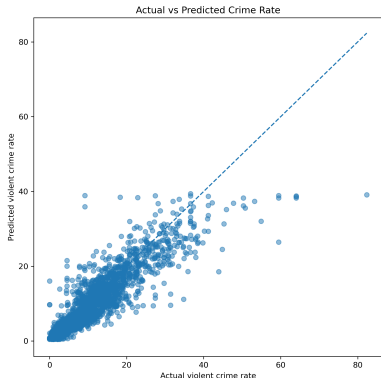
Where does the improvement come from?

	Features	Holdout RMSE	RMSE vs spatial OLS
Spatial Durbin-style OLS	98	3.822	—
XGB same as OLS	99	3.786	−0.94%
XGB expanded pool	269	3.803	−0.50%

Interpretation

- With a similar feature set as OLS, XGBoost has a lower RMSE, suggesting gains from feature interactions and other non-linearities.
- MAE also drops, making it less likely that the RMSE gain is driven only by a few large squared-error outliers.
- Adding 170 additional predictors does not improve RMSE, suggesting that these variables add little extra predictive signal.

LSTM: Stable but Dominated, with Upper-Tail Bias



Prediction cap near 40; high-crime months are under-predicted.

Holdout RMSE, 10 random seeds

Statistic	Value
Mean	3.966
Std	0.060
Min	3.865
Max	4.073
Median	3.979

Seed-level variation is small relative to the gap to OLS, RF, and XGB.

Five mechanisms

- 1 **Strong short-run persistence:** $\hat{\rho}_0 + \hat{\rho}_1 > 0.3$ across all OLS specifications.
- 2 **Fixed effects absorb cross-sectional heterogeneity:** SD of α_i is about 6.5.
- 3 **High sample-to-parameter ratio:** 9,009/98 keeps OLS in a low-variance setting.
- 4 **Near-linearity in crime lags:** most non-linearity appears in the upper tail.
- 5 **Weak marginal signal from auxiliary predictors:** weather, 311, and spatial features each move RMSE only modestly.

In one line: AR + FE already captures the dominant panel structure.

Three takeaways

- 1 OLS with community-area and month FE is a hard baseline:
 $R^2_{\text{oos}} = 0.836$.
- 2 XGBoost on the OLS feature set gives the best forecast; expanding the predictor pool does not improve accuracy.
- 3 All four model families compress the upper tail; LSTM does so most strongly.

Future work:

- Lasso, Ridge, and Elastic Net benchmarks.
- PCA robustness checks on ACS socioeconomic features.
- Cluster bootstrap confidence intervals for paired loss differences.
- Feature-block ablation and upper-tail models.

- 1 Introduction
- 2 Data
- 3 Methods
- 4 Results
- 5 Selected References**

- Becker, G. S. (1968). Crime and Punishment: An Economic Approach. *JPE* 76(2): 169–217.
- Breiman, L. (2001). Random Forests. *Machine Learning* 45(1): 5–32.
- Chen, T. and Guestrin, C. (2016). XGBoost: A Scalable Tree Boosting System. *KDD*.
- Diebold, F. X. and Mariano, R. S. (1995). Comparing Predictive Accuracy. *JBES* 13(3).
- Friedman, J. H. (2001). Greedy Function Approximation: A Gradient Boosting Machine. *Annals of Statistics* 29(5).
- Glaeser, E. L., Sacerdote, B., and Scheinkman, J. A. (1996). Crime and Social Interactions. *QJE* 111(2): 507–548.
- Hochreiter, S. and Schmidhuber, J. (1997). Long Short-Term Memory. *Neural Computation* 9(8).
- Mohler, G. O. et al. (2011). Self-Exciting Point Process Modeling of Crime. *JASA* 106(493).

Thank You